

Javier González Delgado

MATHEMATICIAN - STATISTICIAN

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Professional background

École Nationale de la Statistique et de l'Analyse de l'Information (ENSAI)

Rennes (France)

ASSISTANT PROFESSOR IN STATISTICS

Jan. 2025 - Present

McGill University

Montréal (Canada)

VISITING PROFESSOR

May 2025 - Oct. 2025

POSTDOCTORAL RESEARCHER

Dec. 2023 - Dec. 2024

Research conducted in the Simon Gravel group at the Department of Human Genetics.

International Agency for Research on Cancer (IARC-WHO)

Lyon (France)

MASTER INTERNSHIP

Mar. 2020 - Aug. 2020

Supervised by Vivian Viallon (IARC-WHO), Hadrien Charvat (IARC-WHO) and Grégoire Rey (INSERM).

Academic background

University of Toulouse III - Paul Sabatier

Toulouse (France)

PHD IN APPLIED MATHEMATICS

Oct. 2020 - Oct. 2023

- Manuscript: Statistical methods for the structural analysis of highly flexible proteins. [HAL]
- Supervised by Pierre Neuval at the Toulouse Institute of Mathematics (IMT) and by Juan Cortés at the Laboratory for Analysis and Architecture of Systems (LAAS-CNRS).
- Funded by the LabEx CIMI.

University of Grenoble-Alpes

Grenoble (France)

MASTER IN STATISTICS AND DATA SCIENCE

Sep. 2019 - Aug. 2020

- Grade Point Average: 16.4/20.

University of Oviedo

Oviedo (Spain)

BACHELOR IN MATHEMATICS

Sep. 2014 - Jul. 2019

- Grade Point Average: 8.39/10.
- Bachelor's Thesis in Geometry and Topology of Manifolds.

BACHELOR IN PHYSICS

- Grade Point Average: 8.576/10.
- Bachelor's Thesis in C^* Algebras and their applications to Quantum Mechanics.
- Erasmus+ program at University of Grenoble-Alpes (Sep. 2016 - Jun. 2017).

Publications and communications

PREPRINTS AND SUBMITTED MANUSCRIPTS

- González-Delgado J, Cortés J, Barducci A, Paloni, M: *Evidence for the Progressive Improvement of All-Atom Force Fields in Reproducing Local Conformational Preferences of Flexible Peptides*, 2025, biorXiv:12.22.695171. [biorXiv]
- González-Delgado J, Deronzier M, Cortés J, Neuval P: *Post-clustering Inference under Dependency*, 2023, arXiv:2310.11822. [arXiv]

JOURNAL PAPERS

- [PNAS] González-Delgado J, Mier P, Bernadó P, Neuval P, Cortés J: *The dependence of the amino acid backbone conformation on the translated synonymous codon is not statistically significant*, Proc. Natl. Acad. Sci. U.S.A., 122(24): e2503264122, 2025. [url]

- [[Bioinfo](#)] [González-Delgado J, Bernadó P, Neuviat P, Cortés J: *Weighted families of contact maps to characterize conformational ensembles of \(highly-\)flexible proteins*, Bioinformatics, 40\(11\): btac627, 2024. \[\[url\]\(#\)\]](#)
- [[EJS](#)] [González-Delgado J, González-Sanz A, Cortés J, Neuviat P: *Two-sample goodness-of-fit tests on the flat torus based on Wasserstein distance and their relevance to structural biology*, Electron. J. Statist., 17\(1\): 1547–1586, 2023. \[\[url\]\(#\)\]](#)
- [[JMB](#)] [González-Delgado J, Sagar A, Zanon C, Lindorff-Larsen K, Bernadó P, Neuviat P, Cortés J: *WASCO: A Wasserstein-based statistical tool to compare conformational ensembles of intrinsically disordered proteins*, Journal of Molecular Biology, 435\(14\): 168053, 2023. \[\[url\]\(#\)\]](#)
- [[JSB](#)] [González-Delgado J, Bernadó P, Cortés J, Neuviat P: *Statistical proofs of the interdependence between nearest neighbor effects on polypeptide backbone conformations*. Journal of Structural Biology, 214\(4\): 107907, 2022. \[\[url\]\(#\)\]](#)

SCIENTIFIC DISSEMINATION (IN FRENCH)

- [[MATAPLI](#)] [González-Delgado J: *Statistique pour démêler le désordre des protéines*. MATAPLI, 138, 49-63, 2025. \[\[url\]\(#\)\]](#)

COMMUNICATIONS

Invited talks and lectures

INFERENCE AFTER HUMAN GENETIC CLUSTERING

- MASCOT-NUM: Statistics and uncertainty quantification for environmental and health applications, Rennes, 2026.

WASCO: A WASSERSTEIN-BASED STATISTICAL TOOL TO COMPARE CONFORMATIONAL ENSEMBLES OF INTRINSICALLY DISORDERED PROTEINS

- Lecture and practical session in Algorithms in Structural Bioinformatics (AlgoSB) winter school: “Intrinsic Disorder in Proteins”. Institute of Scientific Studies of Cargèse, Corsica, Nov. 2022.

Selected conferences and seminars

INFERENCE AFTER HUMAN GENETIC CLUSTERING

- SMPGD (Statistical Methods for Post Genomic Data) 2026. Grenoble, Jan. 2026.

POST-CLUSTERING INFERENCE UNDER DEPENDENCE

- 13th International Conference on Multiple Comparison Procedures. Temple University. Philadelphia, Aug. 2025.
- Journées of the SMAI-MAS group (Modélisation Aléatoire et Statistique). Poitiers, Aug. 2024.
- International Seminar on Selective Inference. Online, Feb. 2024.
- IRMAR (Institut de Recherche Mathématique de Rennes) statistics seminar. Rennes, Oct. 2023.

STATISTICAL METHODS FOR THE INVESTIGATION OF HIGHLY-FLEXIBLE PROTEINS

- Centre de Biologie Structurale (CBS) seminar. Montpellier, May 2023.

WASCO: A WASSERSTEIN-BASED STATISTICAL TOOL TO COMPARE CONFORMATIONAL ENSEMBLES OF INTRINSICALLY DISORDERED PROTEINS

- GGMM (French Graphical and Molecular Modelling Group) 2023 - Young Modellers Conference, Toulouse, May 2023.

GOODNESS-OF-FIT TESTS ON THE FLAT TORUS BASED ON WASSERSTEIN DISTANCE AND THEIR RELEVANCE TO STRUCTURAL BIOLOGY

- GdR ISIS meeting on Optimal Transport and Statistical Learning. Institut Henri Poincaré, Paris, Nov. 2021.
- 52th Journées de Statistiques of the French Society of Statistics (SFdS), Côte d’Azur University, Nice, Jun. 2021.

Scientific dissemination

- How to estimate cancer survival (an overview). Dissemination for non-statisticians. IMT PhD seminar. Toulouse, Mar. 2022.

Funded projects

- *Statistics to unravel randomness in Biology*. Allocation d’Installation Scientifique by Rennes Métropole (2025-2027).

Advising

Master students

- Jasmin Neveu (at McGill University, May. 2026 - Aug. 2026). M1 internship. Co-supervised with Simon Gravel.
- Wandrille Buchy (at ENSAI, Jan. 2026 - Jul. 2026). M2 internship. Co-supervised with Juan Cortés.
- Flavie Kolb (at LAAS, Jun. 2022 - Sep. 2022). M1 internship. Co-supervised with Christophe Zanon and Juan Cortés.

Undergraduate students

- Tina Ding (at McGill University, Sep. 2025 - Dec. 2025). Co-supervised with Simon Gravel.
- Supervision of ENSAI students in research initiation projects (2 students, Dec. 2025 – Apr. 2026).

Teaching

ENSAI

Rennes (France)

- Omics data analysis. Lectures (Cours): 24h, Practical works (TP): 24h, 2024 - 2026.
- Unsupervised classification. Lectures (Cours): 21h, Tutorials (TD): 12h, 2024 - 2026.

Toulouse III - Paul Sabatier University

Toulouse (France)

BACHELOR IN MATHEMATICS

- Set theory I. Tutorials (TD): 28h, 2022 - 2023.
- Linear Algebra I. Lectures (Cours): 28h, Lectures and tutorials (Cours-TD): 64h. 2020 - 2023.
- Mathematics I. Tutorials (TD): 60h, 2020 - 2022.
- Statistical Inference. Practical works (TP): 25h, 2021 - 2023.

Other activities

Review activities

- **For journals in Mathematics/Statistics:** Journal of the American Statistical Association (2024), Journal of Multivariate Analysis (2024), Biostatistics (2024), Electronic Journal of Statistics (2022).
- **For journals in Biology:** Protein Science (2023).

Conference organization

- *Statistics, exposome and environmental health*. Rennes, Jun. 2026 (organizing committee) [[url](#)].

Local scientific animation

- Training of *classes préparatoires* teachers (Stages Lisses) at ENSAI, Apr. 2026.
- Co-organizer of the IMT PhD seminar. Sep. 2021 - Jun. 2023. With Viviana Grasselli and Corentin Kilque.
- Co-organizer of the IMT PhD Statistics and Optimization seminar. Sep. 2021 - Jun. 2023. With Dominique Bontemps.
- Co-manager of *Les maths à Toulouse* Twitter account (IMT, EUR MINT and Labex CIMI). Sep. 2021 - Jun. 2023. With Viviana Grasselli and Isabelle Guichard.

Awards

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| 2023 | PhD thesis award , Prix de thèse Jacques Neveu from the Société de Mathématiques Appliquées et Industrielles (SMAI) - MAS group. | France |
| 2014 | Awarded , Secondary school distinction award. 9.94/10. | Avilés (Spain) |
| 2014 | Awarded , Merit-based grant after University entrance examination. 13.464/14. | Asturias (Spain) |

Language skills ---

- Spanish** Mother tongue
- English** C1 level certificate (CEFR)
- French** C1 level certificate (CEFR)